

Atharva Drapude

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PROFILE

AI Data Science undergraduate with hands-on experience developing Machine Learning, Computer Vision, and Generative AI applications. Proficient in Python, PyTorch, TensorFlow, YOLOv8, OpenCV, Docker, and modern ML workflows from data acquisition to deployment. Built scalable AI solutions for surveillance, fraud detection, and legal document intelligence using Computer Vision, RAG, and LLM-based architectures.

EDUCATION

B.E. in Artificial Intelligence and Data Science

DY Patil College of Engineering, Akurdi

2023–2027

SKILLS

- **Programming:** Python, C++, SQL
- **Machine Learning:** PyTorch, Scikit-Learn, Model Training, Model Evaluation, Feature Engineering, Deep Learning.
- **Computer Vision:** YOLOv8, OpenCV, Object Detection, Image Processing.
- **Generative AI:** RAG, LLMs, Prompt Engineering, FAISS, Transformer Embeddings
- **Data Science:** Data Analysis, Data Visualization, Pandas, NumPy, Matplotlib
- **Tools:** Git, Linux, Docker, FastAPI, REST APIs, MLflow, MS Office 365
- **Core Strengths:** Problem Solving, Workflow Design, Debugging, Documentation, Team Leadership, Stakeholder Communication

INTERNSHIP EXPERIENCE

Lead AI Engineer / Head of AI Division — Trinetra Aerospace Technology

2026

- Led AI strategy discussions with stakeholders to translate business requirements into scalable technical solutions.
- Led development of an AI-powered surveillance platform covering data collection, model training, deployment, and production monitoring.
- Drove performance improvements through testing, optimization, and Explainable AI (XAI) integration.
- Delivered production support and user training while contributing to securing a 2.5 lakh grant.

PROJECTS

AI-Powered Drone Surveillance Object Detection System (SIH 2025 Winner) November 2025 – December 2025

- Developed an AI-assisted drone surveillance solution integrating YOLOv8 and algorithms for large-scale monitoring
- Built an end-to-end pipeline covering dataset preparation, augmentation, training, inference, and evaluation on 3,000+ annotated images.
- Designed a multi-phase training strategy to improve detection robustness across varying altitudes and object scales.

RAG AI Legal Assistant

- Developed a RAG-based legal assistant using Hugging Face LLMs for contextual question answering.
- Processed and indexed 1,000+ legal documents into 10,000+ semantic chunks using transformer embeddings.
- Implemented FAISS-based semantic search for efficient information retrieval.
- Optimized response quality through prompt engineering and context-grounded generation.
- Designed a scalable retrieval pipeline for rapid legal information extraction and document analysis.

POSITIONS OF RESPONSIBILITY

AIML Lead– College Technical Club (GDGC,IASC)

- Led the AI/ML domain by mentoring members and guiding them in machine learning, deep learning, and data science concepts.

CERTIFICATIONS

- Career Essentials in Data Analysis– Microsoft LinkedIn Learning

EXTRACURRICULAR ACTIVITIES

- **Winner – Smart India Hackathon (AICTE) 2025**, selected from 500+ teams.
- **2nd Winner – EntrepreneX Hackathon at JSPM Wagholi, with prize money of 2.51 lakhs**, selected from 100+ teams.
- **1st Winner – Fish Tank held at PCCOER, Ravet, with prize money**, selected from 50+ teams.
- **2nd Winner – Startup Hunt held at IIIT,Pune**, selected from 50+ teams.